

Derivative and multiplicity loci of elementary symmetric polynomials in positive characteristic

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Abstract

We determine the dimensions of all derivative loci and all higher multiplicity loci of an elementary symmetric polynomial over an algebraically closed field of positive characteristic. For the degree- d polynomial in n variables, set $Q = p^{v_p(n-d+1)}$. The common zero locus of its order- q derivatives has dimension $d - q - 1 + \mathbf{1}_{Q > d-q}$, whereas the locus of zeros of multiplicity at least s has dimension $d - s + \mathbf{1}_{Q > d}$. The order-two case settles the dimension question in a conjecture of Orzel, with a necessary correction when d is a power of p . We classify all components attaining the larger dimension and prove that they are generically smooth. Along the coordinate components in the smaller-dimension cases, we determine the local rings and obtain generic multiplicity exactly Q . Finally, we identify the full translation stabilizer scheme and derive a Frobenius normal form. These results distinguish dimension jumps from infinitesimal thickening and give a uniform description of their dependence on the characteristic.

Keywords. Elementary symmetric polynomial; derivative locus; multiplicity locus; positive characteristic; translation stabilizer.

1 Introduction

The geometry of derivative loci of elementary symmetric polynomials has a marked dependence on the characteristic. In characteristic zero, the critical locus of the degree- d elementary symmetric polynomial has dimension $d - 2$. In positive characteristic, the dimension can increase by one; moreover, vanishing of the gradient need not imply vanishing of the polynomial. An exact description must therefore distinguish the critical scheme from the singular scheme of the zero hypersurface. It must also account for nonreduced structure, which can persist even when the dimension does not increase.

This paper determines the dimensions for every derivative order and every order of vanishing. The answer depends on the single integer $p^{v_p(n-d+1)}$. The same integer measures the generic thickness of the coordinate components and determines the translation stabilizer. Thus the characteristic-dependent exceptions admit both an exact numerical criterion and a scheme-theoretic explanation.

1.1 Previous work and the dimension problem

In characteristic zero, Limaye, Mittal, and Pareek [2, Lemma 13] prove the upper bound $d - 2$ for the critical locus. Coordinate subspaces give the matching lower bound; see also the historical discussion in [3]. A general method for bounding singular loci of symmetric hypersurfaces by partition spaces appears in Cafure, Matera, and Privitelli [1, Theorem 3.1 and Remark 3.2]. In particular, their

argument bounds the number of distinct coordinate values and shows that a component attaining the resulting dimension bound must be an entire partition space.

For elementary symmetric polynomials in positive characteristic, Orzel [3] proves that the order-two zero locus has dimension between $d - 2$ and $d - 1$ and proposes a complete criterion in Conjecture 2.11. This question arises in the study of formula complexity and linear projections of elementary symmetric polynomials. The first-order result below replaces the two possible dimensions by a necessary and sufficient arithmetic criterion. It also identifies a prime-power boundary case in which the criterion printed in that conjecture, and in the associated sufficient condition, must be amended.

The contribution goes beyond this first-order question. We determine the full numerical profile of higher derivative and multiplicity loci, classify all components responsible for an increase in dimension, and compute the local schemes along coordinate subspaces. The partition-space dimension principle is inherited from earlier work; the exact congruence criteria and the higher-order and local scheme calculations are the substance of the results proved here.

1.2 Notation and dimension formulas

Let K be an algebraically closed field, and write

$$e_d = e_d(x_1, \dots, x_n) = \sum_{\substack{I \subseteq [n] \\ |I|=d}} \prod_{i \in I} x_i, \quad 2 \leq d \leq n.$$

We write $[n] = \{1, \dots, n\}$ and use the conventions $e_0 = 1$ and $e_j = 0$ when j exceeds the number of variables. For $I \subseteq [n]$, let ∂_I denote differentiation once in each variable indexed by I , with $\partial_\emptyset e_d = e_d$. Since e_d is multilinear, these squarefree derivatives agree with the corresponding Hasse derivatives, and derivatives with a repeated index vanish. Define closed subschemes of $\mathbb{A}_K^n$ by

$$C_q(e_d) = V(\partial_I e_d : |I| = q), \quad 1 \leq q \leq d - 1, \quad (1)$$

$$Z_s(e_d) = V(\partial_I e_d : |I| < s), \quad 2 \leq s \leq d. \quad (2)$$

Thus $C_1(e_d) = \text{Crit}(e_d)$, while $Z_2(e_d)$ is defined by e_d and its first derivatives. More generally, the underlying set of $Z_s(e_d)$ consists of the points at which e_d has order of vanishing at least s . Dimensions refer to underlying closed sets; local rings and multiplicities refer to the displayed defining ideals. The symbol $\mathbf{1}_A$ is 1 when the condition A holds and 0 otherwise.

Theorem 1.1 (Exact dimensions). *Let K be algebraically closed of characteristic $p > 0$, and set*

$$L = n - d + 1, \quad Q = p^{v_p(L)},$$

where $v_p(L)$ is the exponent of p in the prime factorization of L . Then

$$\dim C_q(e_d) = d - q - 1 + \mathbf{1}_{Q > d - q}, \quad 1 \leq q \leq d - 1, \quad (3)$$

$$\dim Z_s(e_d) = d - s + \mathbf{1}_{Q > d}, \quad 2 \leq s \leq d. \quad (4)$$

In characteristic zero the dimensions are $d - q - 1$ and $d - s$, respectively.

In positive characteristic we retain the notation L and Q of Theorem 1.1 throughout. The first-order specialization has the following precise form.

Corollary 1.2 (The order-two criterion). *Let P_- be the smallest p -power at least d , and let P_+ be the smallest p -power strictly greater than d . Then*

$$\dim C_1(e_d) = d - 1 \iff P_- \mid n - d + 1, \quad \dim Z_2(e_d) = d - 1 \iff P_+ \mid n - d + 1.$$

In each case the dimension is $d - 2$ when the divisibility fails.

Proof. The inequalities $Q > d - 1$ and $Q > d$ are equivalent to the two divisibility conditions, respectively. $\square$

The two thresholds coincide unless d is a p -power. If $d = p^a$ with $a \geq 1$, they are d and pd . This distinction is exactly the correction needed in [3, Conjecture 2.11]; a detailed comparison and explicit examples are given in Section 6.

1.3 Components and local structure

A partition space is the linear subspace on which coordinates agree within each block of a set partition of $[n]$. We call a component *extremal* when its dimension equals the larger of the two values in the corresponding formula.

Theorem 1.3 (Extremal components). *In positive characteristic all extremal components are as follows.*

1. *For $C_1(e_d)$, partition $[n]$ into $d - 1$ blocks, each of size congruent to 1 modulo the smallest p -power at least d . For $Z_2(e_d)$, use instead the smallest p -power strictly greater than d . The associated partition spaces are exactly the extremal components.*
2. *For $C_q(e_d)$ with $q \geq 2$, put $r = d - q$. When $Q > r$, the extremal components are the spaces on which all but $r - 1$ coordinates are equal. There are $\binom{n}{r-1}$ such components.*
3. *For $Z_s(e_d)$ with $s \geq 3$ and $Q > d$, the extremal components are the spaces on which all but $d - s$ coordinates are equal. There are $\binom{n}{d-s}$ such components.*

Every extremal component is generically smooth in the scheme defined by the relevant derivative ideal, and has multiplicity one.

The smaller-dimension cases can have nonreduced coordinate components. For $S \subseteq [n]$, write $V_S = V(x_j : j \notin S)$ and $F_S = K(x_i : i \in S)$. Generic multiplicity means the length of the local ring at the generic point of an irreducible component.

Theorem 1.4 (Local rings along coordinate subspaces). *Fix $1 \leq q < d$, put $r = d - q$, and let η_S be the generic point of V_S , where $|S| = r - 1$. In characteristic $p > 0$,*

$$\mathcal{O}_{C_q(e_d), \eta_S} \cong \begin{cases} F_S[t]/(t^Q), & Q \leq r, \\ F_S[t]_{(t)}, & Q > r, \end{cases}$$

$$\mathcal{O}_{Z_{q+1}(e_d), \eta_S} \cong \begin{cases} F_S[t]/(t^Q), & Q \leq d, \\ F_S[t]_{(t)}, & Q > d. \end{cases}$$

In each Artinian case, V_S is a component with generic multiplicity Q . In each other case it lies in a component of dimension r . In characteristic zero both local rings are F_S .

Theorems 1.1–1.4 separate two effects of positive characteristic. The inequality $Q > d$ increases the dimension of every Z_s by one. When $1 < Q \leq d$, these dimensions remain at their characteristic-zero values, but the coordinate components have generic multiplicity Q . Dimension alone therefore does not detect the characteristic-dependent scheme structure. The extremal component classification does not purport to describe all smaller-dimensional or embedded components.

1.4 Translation symmetry and the structure of the proof

Proposition 5.1 identifies the full translation stabilizer of e_d with its highest multiplicity scheme $Z_d(e_d)$. It is the diagonal infinitesimal group α_Q when $Q \leq d$, and the diagonal additive group when $Q > d$. In particular, the dimension increase in every Z_s occurs precisely when the translation stabilizer is positive-dimensional. A Frobenius normal form explains why the gradient and the polynomial have different thresholds when $Q = d$.

The proof proceeds from partition spaces to local rings. A common monic polynomial bounds the number of distinct coordinate values at a critical point. On a partition space of the largest possible dimension, the derivative equations become a truncated generating-function identity. Frobenius then converts this identity into exact divisibility conditions on the block sizes. For higher derivatives, comparing two choices of deleted coordinates forces all but one block to be singletons. Finally, localizing a difference of two derivatives identifies the transverse local ring with a quotient of a one-variable discrete valuation ring; its least nonzero order is exactly Q .

Sections 2 and 3 prove the dimension and component statements. Section 4 establishes the local rings and generic smoothness. Section 5 treats translation stabilizers and gradient fibers, and Section 6 gives the conjecture comparison and examples.

2 The first derivative and order-two loci

We first give the elementary block argument underlying the dimension formulas. The bound on the number of distinct coordinates is closely related to the arguments in [1, Theorem 3.1] and [3, Lemma 2.4].

Lemma 2.1. *Every point of $C_1(e_d)$ has at most $d - 1$ distinct coordinate values. Consequently both $C_1(e_d)$ and $Z_2(e_d)$ have dimension at most $d - 1$.*

Proof. The deletion identity gives

$$\partial_i e_d = \sum_{j=0}^{d-1} (-x_i)^j e_{d-1-j}(x).$$

At a critical point all its coordinates are roots of the same polynomial of degree $d - 1$, whose leading coefficient is $(-1)^{d-1}$. There are at most $d - 1$ distinct roots. The critical locus is therefore contained in the finite union of partition spaces with $d - 1$ nonempty blocks. A point with fewer distinct values also lies in one of these spaces, by splitting blocks. $\square$

Lemma 2.2. *Put $r = d - 1$, and let D be a partition space with block sizes $m_1, \dots, m_r$. In characteristic $p > 0$, let P_- and P_+ be the smallest p -powers at least d and strictly greater than d , respectively. Then*

$$D \subseteq C_1(e_d) \iff m_j \equiv 1 \pmod{P_-} \text{ for all } j,$$

and

$$D \subseteq Z_2(e_d) \iff m_j \equiv 1 \pmod{P_+} \text{ for all } j.$$

In characteristic zero either containment would require every m_j to equal 1.

Proof. Use independent block coordinates $y_1, \dots, y_r$, and set

$$E(T) = \prod_{j=1}^r (1 + y_j T)^{m_j}, \quad R(T) = \prod_{j=1}^r (1 + y_j T), \quad H(T) = \frac{E(T)}{R(T)}.$$

If $D \subseteq C_1(e_d)$, the polynomial

$$\sum_{j=0}^r (-U)^j e_{r-j} |_D$$

has the r distinct roots $y_1, \dots, y_r$ over $K(y_1, \dots, y_r)$. It must equal $(-1)^r \prod_j (U - y_j)$. Comparing coefficients, and conversely using the same identity, gives

$$D \subseteq C_1(e_d) \iff H(T) \equiv 1 \pmod{T^d}. \quad (5)$$

Under this congruence, the coefficient of T^d in E vanishes exactly when that in H vanishes, because R has degree $d - 1$. Thus

$$D \subseteq Z_2(e_d) \iff H(T) \equiv 1 \pmod{T^{d+1}}. \quad (6)$$

If a positive integer M has p -adic valuation a , the first nonconstant term of $(1 + z)^M$ in characteristic p has degree p^a . Indeed, writing $M = p^a b$ with $p \nmid b$ gives $(1 + z)^M = (1 + z^{p^a})^b$, with nonzero coefficient b in that degree. It follows that

$$\prod_j (1 + y_j T)^{m_j - 1} \equiv 1 \pmod{T^h}$$

if and only if each $m_j - 1$ is divisible by the smallest p -power at least h . Necessity follows by setting all but one y_j to zero, and sufficiency follows by multiplication. This proves the claims. In characteristic zero the coefficient of $y_j T$ already forces $m_j - 1 = 0$. $\square$

Both loci contain every coordinate space supported on at most $d - 2$ indices. Their dimensions therefore lie between $d - 2$ and $d - 1$. By Lemma 2.1, an irreducible component of dimension $d - 1$ must be an entire partition space with $d - 1$ blocks: a proper closed subset of such a space has smaller dimension. There exists a partition into r blocks of sizes 1 modulo P if and only if $P \mid n - r$. For sufficiency, take $r - 1$ singleton blocks and one block of size $1 + n - r$. Lemma 2.2 now proves Theorem 1.1 for $q = 1$ and $s = 2$, and the first part of Theorem 1.3. It also proves the characteristic-zero dimensions, since $n \geq d > r$ rules out a partition consisting only of singleton blocks.

3 Higher derivatives and higher multiplicity

Fix $2 \leq q \leq d - 1$, and put $r = d - q$. A point of $C_q(e_d)$ has at most r distinct coordinate values. Otherwise choose $r + 1$ coordinates with distinct values and $q - 1$ further indices B disjoint from them; this is possible since $n \geq r + q$. The polynomial $\partial_B e_d$ is elementary symmetric of degree $r + 1$ in the remaining variables, and its gradient vanishes at the point. Lemma 2.1 gives a contradiction.

Thus C_q is contained in the finite union of partition spaces with r blocks. Both C_q and Z_{q+1} contain the coordinate spaces supported on $r - 1$ indices. Their dimensions lie between $r - 1$ and r ; an r -dimensional component must be a full partition space of this size.

Lemma 3.1. *In positive characteristic, let P_r be the smallest p -power strictly greater than r . A partition space with r blocks lies in $C_q(e_d)$ if and only if it has $r - 1$ singleton blocks and one block of size $L + q$, with $P_r \mid L$.*

Proof. Suppose the block sizes are $m_1, \dots, m_r$. Remove $q - 1$ indices without emptying any block, say b_j indices from block j . Such a removal is possible because $\sum_j (m_j - 1) = n - r \geq q$. For

every such removal, the gradient of the remaining elementary polynomial of degree $r + 1$ vanishes identically on the partition space. By Lemma 2.2,

$$m_j - b_j \equiv 1 \pmod{P_r}, \quad 0 \leq b_j \leq m_j - 1, \quad \sum_j b_j = q - 1. \quad (7)$$

The capacities $c_j = m_j - 1$ sum to $n - r \geq q$. If at least two are positive, a feasible vector b of total $q - 1$ can be changed by transferring one unit between two distinct coordinates. To see this, there is a positive entry of b and an unfilled capacity, because $0 < q - 1 < \sum c_j$. Unless a transfer is possible, both sets of indices must be the same singleton; all other capacities would then be zero. The transfer contradicts (7). Hence exactly one block is nonsingleton. Its size is $M = L + q$. Removing $q - 1$ indices from it shows that $P_r \mid L$.

Conversely, suppose these conditions hold. Differentiate with respect to q distinct variables, taking a of them from singleton blocks and $q - a$ from the large block. Denote the common value on the large block by b and the remaining singleton values by $y_1, \dots, y_{r-1-a}$. The restricted derivative is the coefficient of T^r in

$$\prod_{j=1}^{r-1-a} (1 + y_j T)(1 + bT)^{L+a}.$$

Modulo T^{r+1} , replace the last exponent by a , since $P_r > r$ and $P_r \mid L$. The resulting polynomial has degree $(r - 1 - a) + a = r - 1$. Its coefficient of T^r is zero. $\square$

In characteristic zero the same removal argument would require every remaining block size to be 1. Their sum is $n - q + 1 \geq r + 1$, which is impossible. Lemma 3.1 therefore proves (3), the characteristic-zero case, and the component description for C_q .

For Z_{q+1} , let P be the smallest p -power strictly greater than d . An extremal partition already has the shape in Lemma 3.1 and satisfies $P_r \mid L$. It remains to show that $P \mid L$. If $P_r > d$, then $P_r = P$: the inequality $r < d$ gives $P_r \leq P$, and $P_r > d$ gives $P \leq P_r$ by the minimality of P . Thus $P \mid L$ already holds in this case.

Now suppose $P_r \leq d$. Let R be a p -power with $r < R \leq d$ and $R \mid L$. Differentiate with respect to $h = d - R$ variables, all in the large block. This is an allowed order since $0 \leq h < q$. The coefficient of the pure monomial b^R in the restricted derivative is

$$\binom{L + R - r}{R} \equiv L/R \pmod{p}.$$

Here $0 < R - r < R$; the congruence follows by expanding $(1 + z)^{L+R-r} = (1 + z^R)^{L/R}(1 + z)^{R-r}$ in characteristic p . Vanishing forces $pR \mid L$. Starting at $R = P_r$ and applying this step while $R \leq d$ proves divisibility by the next p -power at each stage. The first p -power greater than d is P , so the iteration gives $P \mid L$.

Conversely, assume $P \mid L$. For a derivative of order $0 \leq h \leq q$, let a be the number of differentiated variables from singleton blocks. Write the remaining singleton values as $y_1, \dots, y_{r-1-a}$. Its generating polynomial is

$$\prod_{j=1}^{r-1-a} (1 + y_j T)(1 + bT)^{L+q-h+a}.$$

Modulo T^P , replace the last exponent by $q - h + a$. The resulting degree is at most $d - h - 1$, while the required coefficient has degree $d - h < P$. It vanishes. This proves (4) and the remaining component descriptions. The number of these partition spaces is the number of choices of singleton coordinates, as stated.

4 Local multiplicities and generic smoothness

4.1 Coordinate components

We prove Theorem 1.4. Put $B = [n] \setminus S$, so $|B| = M = L + q$. Work in the local ring of $\mathbb{A}^n$ at $(x_j : j \in B)$. For $i, j \in B$, choose $q - 1$ other indices $T \subset B \setminus \{i, j\}$. The identity

$$\partial_{T \cup \{i\}} e_d - \partial_{T \cup \{j\}} e_d = (x_j - x_i) \partial_{T \cup \{i, j\}} e_d \quad (8)$$

has a unit as its last factor: $|B| = L + q \geq q + 1$ guarantees the choice of T , and the restriction of that factor to V_S is $\prod_{a \in S} x_a$. Thus both derivative ideals contain all differences $x_i - x_j$ on B after localization. The ambient local ring is $F_S[x_j : j \in B]_{(x_j : j \in B)}$. Quotienting by these differences and writing their common value as t gives $F_S[t]_{(t)}$. It remains to compute the ideal generated by the restricted derivatives in this discrete valuation ring.

Consider a derivative of order $h \leq q$ removing a indices from S , and put $u = q - h$. Its restriction is the coefficient of T^{d-h} in

$$\prod_{i \in S \setminus A} (1 + x_i T) (1 + tT)^{L+u+a}, \quad |A| = a. \quad (9)$$

A possible power t^j satisfies $u + a + 1 \leq j \leq r + u$. If $j < Q$, then $u + a < Q$, and its binomial coefficient reduces to $\binom{u+a}{j} = 0$ modulo p . Hence every nonzero restricted generator has t -order at least Q .

For the order- q ideal, the derivative using only indices in B restricts to

$$\sum_{j=1}^r \binom{L}{j} t^j e_{r-j}(x_S).$$

When $Q \leq r$, its coefficient of t^Q is the nonzero element $(L/Q) e_{r-Q}(x_S)$ of F_S . The ideal in the discrete valuation ring is therefore (t^Q) . If $Q > r$, all order- q restrictions vanish identically by the block criterion.

For the ideal of Z_{q+1} , this also suffices when $Q \leq r$. If $r < Q \leq d$, use the derivative of order $h = d - Q$, all in B . In (9) the coefficient of t^Q is $\binom{L+Q-r}{Q} = L/Q \neq 0$ in K . If $Q > d$, all restrictions vanish identically, by the preceding section. The dimension formulas identify the component cases, and the length of $F_S[t]/(t^Q)$ is Q . In characteristic zero the coefficient of t in the order- q restriction is $L \prod_{a \in S} x_a \neq 0$, giving the local ring F_S .

4.2 Extremal components

For $q = 1$, take a partition component with distinct block values $y_1, \dots, y_{d-1}$. Extracting the coefficient of degree $d - 2$ after two deletions shows that the Hessian of e_d has zero cross-block entries, and block j equals

$$c_j(J_{m_j} - I_{m_j}), \quad c_j = \prod_{\ell \neq j} (y_\ell - y_j) \neq 0.$$

Here J_m is the all-one matrix and I_m the identity matrix. Since $m_j \equiv 1 \pmod{p}$, the kernel of $J_{m_j} - I_{m_j}$ is exactly the line of constant vectors, so its rank is $m_j - 1$. The Hessian rank is therefore $n - (d - 1)$, exactly the codimension of the partition space. The critical scheme is smooth there. If that space also lies in Z_2 , the latter scheme has the same dimension and is smooth there as well.

For $q \geq 2$, put $r = d - q$ and take an extremal partition with $r - 1$ singleton values a_i and a common value b on the block B of size $M = L + q$. Assume $a_i \neq b$ for all i . In the Jacobian of the order- q equations, use the rows indexed by q -subsets I of B . Their entries at singleton coordinates are zero; the entry at $j \in B$ is

$$\begin{cases} c, & j \notin I, \\ 0, & j \in I, \end{cases} \quad c = \prod_i (a_i - b) \neq 0.$$

The complement-incidence rows have coordinate sum $M - q = L = 0$ in K . For distinct $i, j \in B$, choose a $(q - 1)$ -subset of $B \setminus \{i, j\}$ and compare the two q -subsets obtained by adjoining i and j . The corresponding row difference is a nonzero scalar multiple of $e_i - e_j$. Thus the row space contains, and is contained in, the coordinate-sum-zero subspace on B . Its rank is $M - 1 = n - r$. The scheme contains the r -dimensional partition space, so this is precisely the rank required for smoothness. The same argument applies to Z_{q+1} . This completes the proof of Theorem 1.3.

5 Translation stabilizers and a normal form

For each K -algebra A , consider the translation stabilizer

$$\mathcal{T}(A) = \{v \in A^n : e_d(x + v) = e_d(x) \text{ in } A[x]\}.$$

Write $\alpha_Q = \text{Spec } K[t]/(t^Q)$ when Q is a p -power; this is the kernel of the Q th-power homomorphism of $\mathbb{G}_a$.

Proposition 5.1. *In positive characteristic the translation stabilizer is the diagonal copy of α_Q if $Q \leq d$, and the diagonal copy of $\mathbb{G}_a$ if $Q > d$. In characteristic zero it is the reduced origin. Moreover, $\mathcal{T} = Z_d(e_d)$ as closed subschemes.*

Proof. The coefficient of x_I , for $|I| = d - 1$, in $e_d(x + v) - e_d(x)$ is $\sum_{j \notin I} v_j$. Subtracting the equations for two such subsets differing by one index forces $v_i = v_j$ for every i, j , over any test algebra. Write the common coordinate as t . The remaining equations are

$$\binom{L + h - 1}{h} t^h = 0, \quad 1 \leq h \leq d. \quad (10)$$

If $L = Qu$ with $p \nmid u$, the generating series is

$$(1 - z)^{-L} = (1 - z^Q)^{-u}.$$

Its first nonconstant coefficient has degree Q and equals the unit u ; all its positive degrees are multiples of Q . Thus (10) generates (t^Q) when $Q \leq d$, and the zero ideal when $Q > d$. In characteristic zero the first equation already gives $Lt = 0$.

For a homogeneous multilinear polynomial of degree d , the coefficients of degrees less than d in its Taylor expansion at v are exactly the derivatives of orders less than d at v . Its degree- d part is the original polynomial. Therefore the stabilizer equations define precisely $Z_d(e_d)$. $\square$

In particular, the defining ideal of Z_d in positive characteristic is

$$\begin{cases} (x_1 - x_n, \dots, x_{n-1} - x_n, x_n^Q), & Q \leq d, \\ (x_1 - x_n, \dots, x_{n-1} - x_n), & Q > d. \end{cases}$$

The dimension jump in every Z_s thus occurs exactly when the translation stabilizer has positive dimension. A nontrivial finite infinitesimal stabilizer alone does not produce this jump.

There is also an explicit normal form. Set $z_i = x_i - x_n$ for $i < n$ and $t = x_n$. The translation expansion gives

$$e_d(x) = \sum_{b=0}^{\lfloor d/Q \rfloor} \binom{u+b-1}{b} t^{bQ} e_{d-bQ}(z_1, \dots, z_{n-1}), \quad u = L/Q. \quad (11)$$

Here $e_0 = 1$ and $e_j = 0$ for $j > n-1$. When $Q > d$, the polynomial is independent of t . When $Q = d$, it is $e_d(z) + ut^d$; its gradient is still independent of t . Thus the critical-locus threshold is $Q \geq d$, whereas the order-two zero threshold is $Q > d$. These differ precisely at p -power degrees.

Corollary 5.2. *If $Q \geq d$, every gradient fiber over a target $a = (a_1, \dots, a_n)$ with $\sum_i a_i = 0$ is an affine line times the corresponding gradient fiber of e_d in $n-1$ variables. Targets outside that hyperplane have empty fibers.*

Proof. Let $g = \nabla e_d(z_1, \dots, z_{n-1})$. Differentiating (11), in the original coordinates, gives

$$(g_1, \dots, g_{n-1}, -\sum_i g_i),$$

independently of t . This is an identity of morphisms and hence gives the stated scheme-theoretic product of fibers. $\square$

6 The order-two criterion and examples

Let P_+ be the smallest p -power strictly greater than d . Formula (4) says

$$\dim Z_2(e_d) = d-1 \iff n-d+1 \equiv 0 \pmod{P_+}.$$

Conjecture 2.11 of [3] uses instead $p^{\lceil \log_p d \rceil}$, together with $n \geq 2d-1$. Proposition 2.6 of that revision states a sufficient condition with the same threshold. When d is not a p -power, that threshold is P_+ , and its divisibility condition already implies $n \geq 2d$. When d is a p -power, the strict threshold must instead be pd . Theorem 1.1 proves the necessary and sufficient criterion with this correction.

For the smallest example, take $p = 2, d = 2, n = 3$. The gradient of

$$e_2 = x_1x_2 + x_1x_3 + x_2x_3$$

vanishes on the diagonal (a, a, a) , where $e_2 = a^2$. Hence $\dim C_1(e_2) = 1$ but $Z_2(e_2)$ is supported at the origin. Its coordinate ring is $K[t]/(t^2)$ by Proposition 5.1. The printed conditions $n \geq 2d-1$ and $p^{\lceil \log_p d \rceil} \mid n-d+1$ hold, but $\dim Z_2(e_2) = 0$.

An odd-characteristic example is $p = 3, d = 3, n = 5$. On the partition space (a, b, b, b, b) , every first derivative vanishes, while $e_3 = b^3$. The exact dimensions are $\dim C_1(e_3) = 2$ and $\dim Z_2(e_3) = 1$. Every coordinate-axis component of the latter has generic multiplicity three. More generally, when $d = p^a$ and $n = 2d-1$, the two dimensions are $d-1$ and $d-2$.

The correction concerns the arithmetic criterion for attaining dimension $d-1$. It leaves unchanged the uniform upper bound used in the formula lower-bound argument of [3].

7 Consequences and further questions

The results determine the full dimension profile of the schemes $C_q(e_d)$ and $Z_s(e_d)$ and explain the exceptional dimensions through translation symmetry. Their scheme-theoretic content is independent of a dimension increase: if $1 < Q \leq d$, the coordinate components of Z_s have nontrivial generic thickness even though their dimensions agree with those in characteristic zero. At the other extreme, $Q > d$ gives an actual translation direction and increases every higher multiplicity dimension by one.

For the first-order problem, the distinction is especially explicit. At $Q = d$, the normal form is $e_d(z) + ut^d$. The term ut^d is invisible to ordinary first derivatives, but it remains in the equation of the zero hypersurface. This accounts simultaneously for the difference between C_1 and Z_2 , the strict divisibility threshold in Corollary 1.2, and the infinitesimal stabilizer α_d .

In applications that use derivative-locus codimension, the formulas supply the exact geometric input in every characteristic and degree. The present paper does not derive a new asymptotic formula or circuit lower bound from that refinement. Such an application would require an additional argument connecting the relevant computational model to these exact loci.

Several geometric questions remain. The smaller-dimensional and embedded components are not classified here. Determining primary decompositions or a general criterion for global reducedness would require information beyond the generic coordinate local rings. Likewise, Corollary 5.2 describes all gradient fibers in the range where the gradient is invariant under diagonal translations, but does not classify the fibers in the remaining small-characteristic cases. These questions concern the geometry left after the numerical dimension problem has been resolved.

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